

19/30

Question:— 3

(a) Here is the 3 address code corresponding to the DAG

$$t_1 = a + a$$

$$t_2 = t_1 + t_1$$

$$t_3 = t_2 - t_1$$

$$t_4 = t_3 - t_2$$

(b) The quadruples representation uses a table:

Operator	Arg 1	Arg 2	Result
+	a	a	t_1
+	t_1	t_1	t_2
-	t_2	t_1	t_3
-	t_3	t_2	t_4

Triples Representation:

The triples representation uses a list of triples, where each triple consists of an operator and two operands.

Index	Operator	Arg1	Arg2
(0)	+	a	a
(1)	+	(0)	(0)
(2)	-	(1)	(0)
(3)	-	(2)	(1)

In the triples representation, the index is used instead of a temporary variable to refer to the result of a computation.

Question:-2

① Global scope:-

float x

float v

int m[10][20]

② Main Function scope:

'int argCount'

'char* argValues[]'

'float t'

'int rc'

③ for first for loop scope:

'int i'

'float s'

④ second for loop scope:

'int j'

'int c'

5.5

⑤ innermost while loop:

(int v)

(int count)

(float rc)

(float m)

when the CPU executes the line ' $rc = rc * m;$ ' inside the while loop the active symbol table would only consider variables that are in the current scope or in any parent scope that is currently active

Hash table :

Scope	Name	Type
while loop	count	int
while Loop	rc	float
while Loop	m	float
Forc Loop 2	7	int
ForLoop 2	c	int
Forc loop 2	v	int
Forc Loop 1	s	float
Forc Loop 1	i	int
Main	rc	int
Main	t	float
Main	arg count	int
Main	arg values	Char*[]
Global	m	int [10] [20]
Global	v	float
Global	x	float

(Showing 'rc' in
main)

(Shadowing 'm'
in Global)

(Shadowing 'v'
in Global)

Not a hash table

Question 1-1

The annotated parse tree for ~~font~~ ?

